

A confirmed beta-harvest cannot pass a prop-firm challenge without regime foreknowledge

A pre-registered first-passage falsification on the FTMO two-step, 2010 to 2026

trading2moon

Working paper · July 2026 · deployment-context negative result

Abstract. A proprietary-trading-firm evaluation is not a return problem, it is a first-passage problem: an account equity path must reach an upside target before it ever touches a daily-loss or a maximum-loss barrier. That distinction is why “you do not need alpha, just harvest market beta and size it small” is a tempting but wrong recipe for passing one. We test it directly against the FTMO two-step, using a rule-faithful daily barrier simulator over sixteen years of futures data (2010 to 2026), and the answer is a clean no on three successively more favorable setups. First, the pre-registered, diversified six-leg beta basket (vol-targeted, inverse-vol) tops out at a structural **+1.0%/yr** drift ceiling and fails *even at zero cost*: at a provisional single-slice adjudication its probability of passing and retaining is exactly zero at both the 63-day and 126-day windows and its drift-adjusted edge is zero, with the miss being the profit target and zero loss breaches, so the wall is thin drift, not risk (the full walk-forward gate is pending, but a +1%/yr harvest cannot reach a +15% target regardless of the fold split). Second, a deliberately post-hoc concentration into the two drift-fattest legs, {NQ, GC}, is positive only on a hindsight-declared bull window (business EV **+\$681**), unconditionally a coin-flip around zero (**-\$28** at the pinned vol target, **+\$24** at the higher one), and **-\$489** in the 2022 bear; all of its apparent edge is realized 2023 to 2026 drift, its driftless null stays low (0.04 to 0.06 at the pinned vol target) in every regime. Third, the one thing that could supply the needed foreknowledge without cheating, a look-ahead-free mechanical 200-day moving-average regime gate (a provisional realized-path probe), whipsaws the +10.1%/yr hindsight drift down to **+0.1%/yr** across 27 switches and deepens the drawdown rather than improving it. The contribution is methodological as much as empirical: a driftless null that de-means inside the barrier engine to separate realized drift from geometry, a cost-independence check that isolates the binding wall, and an honest upper-bound probe that shows even maximal hindsight concentration does not clear the fee. The result is consistent with a 0/16 program base rate and does not overturn it; the recipes remain derived and NOT-PROVEN.

1 Introduction

There is a piece of folklore in the proprietary-trading-firm (prop-firm) world that sounds like sober risk management: you do not need a real edge to pass a funded-account challenge, you only need to harvest a bit of market beta, size it conservatively, and let diversification carry you past the profit target while the small size keeps you clear of the loss limits. It is attractive precisely because the first half is true. You genuinely do not need alpha in the sense of a risk-adjusted excess return. What the folklore misses is that a prop-firm evaluation is a *first-passage problem*, not a mean-return problem, and first-passage problems are unforgiving in a way that averages hide.

A first-passage problem asks which of several barriers a random path touches first. Passing the FTMO two-step means an account's equity must climb to a profit target (ten percent in phase one, five percent in phase two) *before* it ever touches a daily-loss barrier (down five percent from the day's starting balance) or an overall maximum-loss barrier (down ten percent). Plain-language version: it is a race between a modest finish line above you and two trap-doors below and beside you, and you have to reach the finish without stepping on a trap-door even once. A strategy with a positive average return can still lose that race almost surely, if its drift is too thin to reach the target inside a deployable window, or if its path variance trips a barrier on the way. This paper measures exactly that, for the specific recipe the folklore prescribes.

We do not report the negative result as a disappointment. A clean, pre-registered, adversarially challenged demonstration that a widely-repeated “just harvest beta” recipe cannot pass a real prop-firm challenge is useful: it tells an operator where not to spend a challenge fee, and it isolates *why* the recipe fails, which turns out to be a structural drift ceiling rather than a risk-management failure. Section 2 describes the data, venue, and cost model; Section 3 the barrier simulator, the driftless null, and the business-EV accounting; Section 4 the three results in increasing order of how favorable we make the setup to the strategy; Section 5 the kill batteries; and Section 6 what this says about beta harvesting under barriers. A firewall note first (Section 1.1), because it governs how these numbers may and may not be used.

1.1 What this is, and what it is not

This is a *deployment-context* write-up, not a research finding. Under the program's one-way-street rule (ADR 0064, section 3), a prop-deployment work package may consume the research corpus but may never write back to it. Accordingly nothing here is recorded to the findings or belief ledgers, and nothing here moves the program's 0/16 research base rate. The three “beta recipes” this round tested remain labeled *derived* and *NOT-PROVEN*, exactly as they were before. What the paper adds is a sharper, barrier-native reason for that standing verdict, and a reusable method for producing it. The concentrated {NQ, GC} result in Section 4.2 is, by explicit construction, a *post-hoc* probe (a HARKing upper bound, defined in Section 3), included to bound the best case, not to propose a strategy.

2 Data and venue

All price paths are built from Databento GLBX.MDP3 daily bars on CME futures, 2010 through 2026. The pre-registered diversified basket has six legs, each a futures proxy for the corresponding FTMO contract-for-difference (CFD): US500 (ES), XAUUSD (GC), EURUSD (6E), USOIL (CL), XAGUSD (SI), and USDJPY (6J), aligned to a common 4,075-day sample. The concentrated probe uses the two US-index and gold legs, NQ and GC, on a common 4,048-day sample. A continuous front-contract series is built per root from the largest-volume outright, with rolls handled so that no return and no indicator window spans a roll gap.

Two honest gaps in the data model, both flagged as pre-registered threats to validity. First, the price paths come from *futures*, while the fees, financing, and fills that an operator would actually pay come from FTMO *CFDs* the basis difference is real and is the reason the cost inputs below are placeholders. Second, we do not have the real FTMO fee schedule, per-symbol swap, or event calendar frozen yet (that is a human-in-the-loop source freeze, deferred), so costs are modeled at a placeholder overnight swap of 3.57 basis points per night on notional, roughly nine percent per year, charged per trading day on both legs. The evaluation-account geometry is a \$100,000 account, a \$540 refundable challenge fee, and an 0.8 payout split. The single most important robustness fact

in the paper, developed in Section 4.1, is that the central verdict does *not* depend on those cost placeholders: the diversified basket fails even at zero cost.

3 Methodology

Four choices do the work of turning a barrier backtest into a verdict. They are the paper’s transferable content, so we state them up front.

3.1 A rule-faithful daily first-passage simulator

The core engine replays the FTMO two-step as a genuine first-passage process on daily (Open, Low, Close) equity triples. Each simulated challenge draws a path by block-bootstrap (20-trading-day blocks, so that autocorrelation and volatility clustering survive the resample), sizes the basket to a target annualized volatility with inverse-volatility leg weights and a monthly rebalance, and then checks, day by day, the real rules: the profit target is measured on *balance* (closed equity), the daily-loss limit is anchored to the balance at the day’s start (not to trailing equity, which would be anti-conservative under an unrealized drawdown), the maximum-loss limit is a static floor, and a minimum-trading-days count must be met. The intraday *Low* of the triple is what tests the daily-loss and max-loss barriers, aggregated across legs under a conservative *co-bottoming* assumption (legs are taken to bottom together on the day, which overstates the worst-case intraday excursion and therefore makes passing harder, not easier). Plain-language version: we do not score the strategy on its average day, we make it run the real obstacle course, thousands of times, and count how often it finishes.

The reported quantity is $P(\text{pass} \wedge \text{retain})$, the joint probability of passing both phases *and* then surviving a pinned 20-trading-day funded horizon without a breach, because a pass that immediately blows the funded account is worth nothing to an operator.

3.2 A driftless null computed inside the engine

A high pass rate is not evidence of skill if the window simply drifted up. To separate the two we compute a *driftless null*. We de-mean each leg’s daily returns inside the evaluation window (subtract the average daily return, keeping the volatility and the shape of the path intact) and then rerun *the entire policy* in that driftless world, producing $P_{\text{driftless}}$. The headline *edge* is the difference, $\text{edge} = P - P_{\text{driftless}}$. Read it carefully: a positive edge does not mean the strategy has alpha, it means the realized window *had drift that the barrier geometry could convert into passes*. Because the de-meaning happens before the policy runs, and the daily stop truncates the path asymmetrically, the null has to be regenerated through the policy rather than by shifting the output, or it would misstate the floor. In every regime we test, the driftless floor stays low, 0.04 to 0.06 (Figure 3): without drift, the barrier geometry alone essentially never passes.

3.3 Business EV, not raw pass probability

The deployment-relevant quantity is not P but the expected dollars per attempt. We compute a *business EV*: the probability of passing and retaining, times the expected retained profit at the 0.8 payout split, minus the \$540 fee, under conservative funded-phase haircuts. A strategy can have a respectable pass probability and still be business-negative once the fee and the haircut chain are paid, which is exactly what happens to the unconditional concentrated basket in Section 4.2.

3.4 Pre-registration and the HARKing guard

The diversified basket, its vol target, its leg set, and the four gates (edge, 2022 survival, business EV, terms-of-service conformance) were all fixed in a frozen pre-registration before any number was produced. The concentrated {NQ, GC} basket was *not*: it was chosen after seeing which legs carried the most realized drift. That is textbook HARKing (Hypothesizing After the Results are

Known, choosing the hypothesis to fit the data you already saw), and we label it as such wherever it appears. We keep the probe only because it is an *upper bound*: if even the maximally cherry-picked concentration is unconditionally negative, the honest pre-registered basket has no hidden rescue.

4 Results

We report three results in increasing order of how favorable we make the setup to the strategy, from the honest pre-registered basket to a deliberately hindsight-optimal concentration.

4.1 The pre-registered harvest hits a structural drift ceiling

The pinned config is the pre-registered one: a six-leg, vol-targeted (nine percent annualized), inverse-vol, monthly-rebalanced long basket, run through the full hazard policy. At a provisional single-slice adjudication (the full gate-grade walk-forward per-fold edge-CI and rolling-2022 origins are still pending) it fails the edge gate outright at both windows: $P(\text{pass} \wedge \text{retain}) = 0$ and $\text{edge} = P - P_{\text{driftless}} = 0$ at 63 days and at 126 days. This is not a barrier problem. At an infinite evaluation window the pass probability is still only 0.02, the dominant reason for not passing is failing to *reach the profit target*, and there are zero daily-loss, drawdown, or minimum-trading-day breaches. The account does not blow up; it simply never arrives.

The cause is a drift ceiling, and it is cost-independent (Figure 1). The vol-targeted inverse-vol construction over-weights the low-drift and negative-drift legs (the FX legs in particular) and dilutes the one strong equity leg down to the nine-percent target, so the basket harvests only about **+1.0%/yr gross**. That is an order of magnitude short of the roughly fifteen percent cumulative a two-step pass requires inside any deployable window, so the pass probability is essentially zero *before any cost is charged*. Adding costs only moves it further from viable: +0.83%/yr at a flat four basis points, and negative (−1.94%/yr and −4.87%/yr) as the placeholder swap is phased in. The pending real-fee freeze can therefore only confirm the STOP, never reverse it. This is the most decisive result in the round precisely because it does not lean on the cost placeholders we are least sure about.

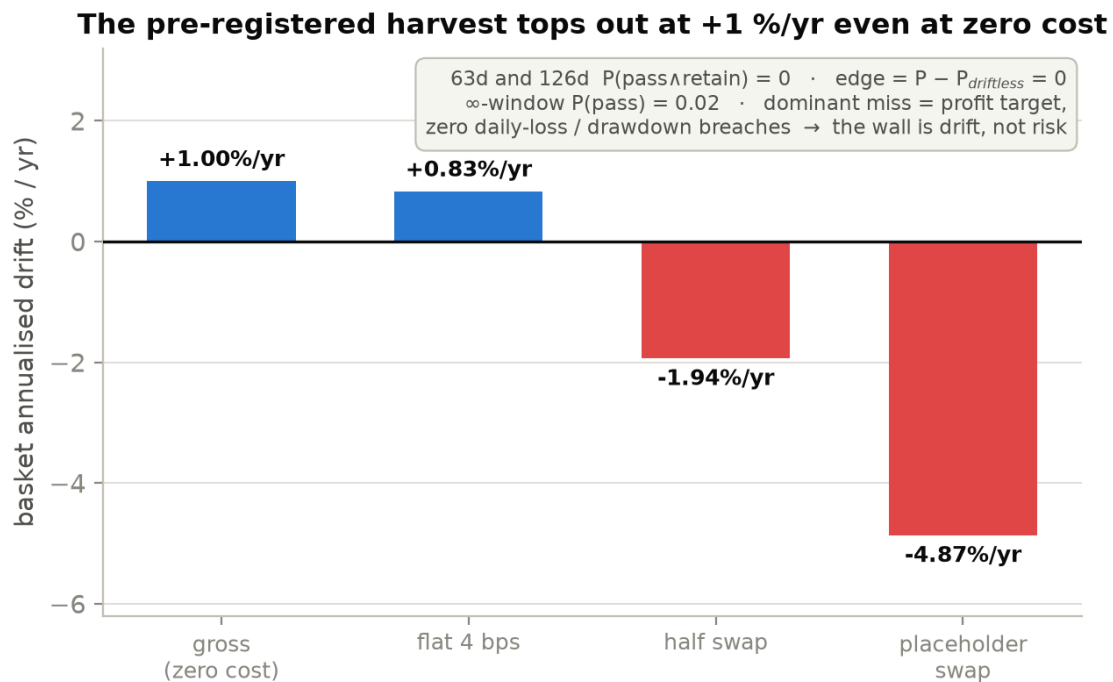


Figure 1: The pre-registered diversified basket’s annualized drift under four cost scenarios. Even gross of cost it harvests only +1.0%/yr, an order of magnitude below what a +15% two-step target needs in a deployable window, so at 63 and 126 days $P(\text{pass} \wedge \text{retain}) = 0$ and the drift-adjusted edge is zero. The dominant failure mode is missing the profit target, with zero loss breaches: the binding wall is thin drift, not risk.

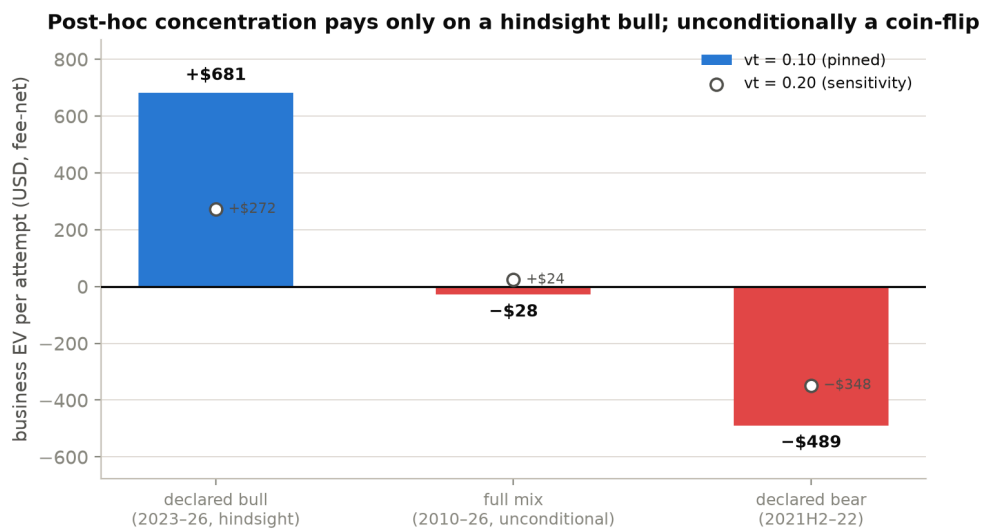
The survival gate, by contrast, passes. On the realized deterministic path the basket ends 2022 at −4.1% with a worst intraday drawdown of −6.1% and a minimum equity of −4.2%, clear of the −10% static max-loss barrier, and the full 2010 to 2026 path ends +8.5% with a worst drawdown of −6.3%. Diversification plus the hazard policy genuinely solved the survival vise that broke the earlier single-asset pins. The binding constraint has simply moved: from “the path blows up in the bear” to “the path never reaches the target.” Both point STOP; the drift ceiling is the more informative of the two because it is structural.

4.2 Concentration is positive only on a hindsight bull, and a coin-flip unconditionally

If the honest diversified basket is too thin to win, the obvious next question, and the one an operator actually asks, is whether concentrating into the strongest legs rescues it. We answer it with the most favorable concentration we can construct after the fact: substitute the higher-drift NQ for the basket’s US500 (ES) leg and drop to just {NQ, GC} (US tech index plus gold), at a slightly higher ten-percent vol target, plain long hold. That is a re-selection, not a subset, so the post-hoc freedom is if anything larger than a pure cherry-pick; this is HARKing by design (Section 3), and it is an upper bound, not a proposal.

Even so, the unconditional bet does not clear the fee (Figure 2). Over the full 2010 to 2026 sample the business EV is **−\$28** per attempt at the pinned ten-percent vol target, a coin-flip around zero. It is strongly positive, **+\$681**, *only* on a hindsight-declared 2023 to 2026 bull window, and **−\$489** in the 2021H2 to 2022 bear, where gold did not hedge the rate shock and both legs fell together. The entire positive expectation lives in knowing the regime in advance. The twenty-percent vol-target sensitivity shows the same pattern, muted: the magnitudes compress toward zero (bull +\$272, bear −\$348, full mix +\$24) because a higher vol target trips more barriers, so the bull-to-bear spread narrows rather than widening. The unconditional EV straddles zero (−\$28 at ten percent, +\$24 at

twenty), well inside Monte-Carlo noise; scaling the size does not manufacture an unconditional edge, it just moves a near-zero number across zero.



The concentrated {NQ, GC} basket is chosen post-hoc (the two drift-fattest legs); it is an upper bound, not a strategy. Unconditionally it is a coin-flip around zero.

Figure 2: Business EV per attempt (fee-net, USD) for the post-hoc {NQ, GC} concentration, by regime. Positive only on the hindsight-declared bull; unconditionally a coin-flip around zero and outright negative in the bear. The open markers are the twenty-percent vol-target sensitivity. The bet is a directional wager on the operator’s own regime call, not a harvestable edge.

The mechanism is visible in the pass probabilities against the driftless null (Figure 3). The driftless floor is low in every regime, 0.04 to 0.06: strip the drift out and the barrier geometry alone almost never passes. The bull’s headline 0.76 pass rate is therefore almost entirely realized 2023 to 2026 drift (edge +0.73 over a 0.04 floor), the full-sample 0.31 is drift too (edge +0.28) but not enough of it to beat the fee, and the bear actually passes *below* its own driftless floor (edge –0.03). A high edge here is a statement about the past window, not a property that repeats out of sample.

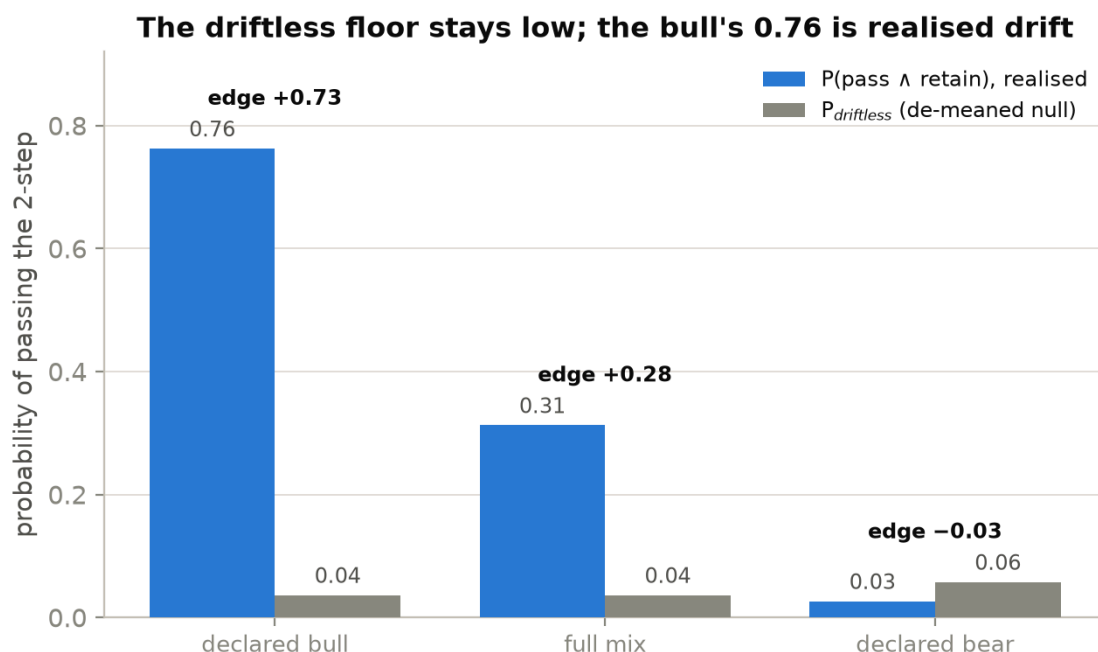


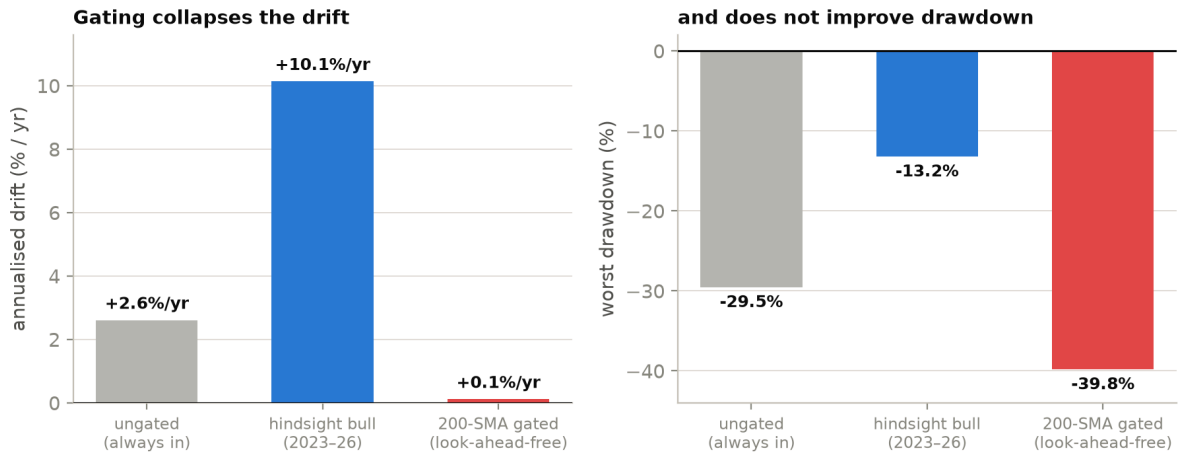
Figure 3: Realized pass probability against the de-meanned driftless null, per regime, {NQ, GC} at a ten-percent vol target. The floor stays low (0.04 to 0.06) across regimes, so the bull's 0.76 is realised drift rather than a repeatable edge. Passing above the null requires the window to have drifted; the null is what a driftless version of the same path, same volatility, would have passed at.

4.3 A mechanical regime gate cannot supply the foreknowledge

The concentrated bet only works if the operator knows the regime in advance, so the last question is whether a rule can supply that knowledge without cheating. We test the simplest honest candidate: a look-ahead-free 200-day simple-moving-average (SMA) gate on NQ, evaluated at each monthly rebalance and held for the month (long the basket when price is above its 200-day average, otherwise flat). This is a *provisional* realized-path probe (realized additive equity paths, not the full barrier engine; see Section 7), and it is the one place we replace a hindsight-declared window with a genuinely causal decision.

The gate gets the direction right and still destroys the edge (Figure 4). It is in the market about 79% of the time, correctly in during the 2023 to 2026 bull and mostly out in 2022 (24% in-market in the bear), not inverted. But the 27 regime switches over sixteen years each forfeit the whipsaw round-trip, exiting after a drop and re-entering after a rise, so that even with the big calls right the opportunity cost grinds the hindsight bull's +10.1%/yr drift down to **+0.1%/yr**, below even the always-in +2.6%/yr, while the maximum drawdown gets *worse*, -39.8% gated against -29.5% always-in. Getting the big calls right is not enough when a lagging trend filter shreds the return on the small ones. The +\$681 bull EV lived in the *foreknowledge* of the regime windows; a causal proxy for that foreknowledge annihilates it.

A look-ahead-free 200-SMA gate whipsaws the edge to zero



27 switches over 16 years (79% in market): direction is clean (in during the bull, mostly out in 2022) but the whipsaw destroys the +10.1 %/yr hindsight drift.

Figure 4: A look-ahead-free monthly 200-day-SMA gate on the {NQ, GC} basket (ten-percent vol target). Left: annualized drift collapses from the +10.1%/yr hindsight bull to +0.1%/yr once the gate is causal, below even always-in. Right: the gate does not improve drawdown, it deepens it. The direction is clean (in during the bull, mostly out in 2022) but 27 whipsaw switches destroy the edge. Provisional: realized additive paths, not the barrier engine (Section 7).

5 Robustness and kill batteries

Each result is stress-tested, and the verdict must survive all of it. The drift-ceiling result (Section 4.1) carries a *cost-independence* check: it fails at zero cost, so no cost placeholder can rescue it, and the failure mode is diagnosed (target-miss, not breach) rather than assumed. The concentration result (Section 4.2) carries the *driftless null* as its central kill: the edge is shown to be realized drift by regenerating the null through the full policy, and the business EV is computed fee-net and haircut-adjusted rather than as a raw pass rate. A *regime split* (bull, full, bear) is reported for every number, which is what exposes the whole bet as conditional on foreknowledge. The gate result (Section 4.3) is a *HARKing antidote*: it replaces the hindsight-declared window with a causal rule and shows the edge does not survive the substitution. A HARKing guard runs throughout, in that the one basket that was pre-registered is kept separate from the one that was cherry-picked, and the cherry-picked one is labeled an upper bound rather than a proposal.

Every number in the concentration probe was additionally put through an independent, adversarial, cross-implementation challenge tasked two-sided with finding either a leak that invents an edge or a bug that suppresses a real one. The review returned *sound with caveats*: the pass-probability and driftless-null pipeline reproduced bit-for-bit, and the “beta not alpha” reading was confirmed (the bull edge is realized drift, driftless floor 0.036, unconditional EV negative). It surfaced one latent code-hygiene defect in a re-origin helper that builds an equity point without its realized-flag, which is immaterial and conservative in this run but is flagged as a live-deployment landmine to fix before reuse. No defect touched a verdict.

6 Discussion

The consistent picture across sixteen years and three successively more favorable setups is that the “just harvest beta” recipe fails a prop-firm challenge for a structural, first-passage reason, not a risk-control one. A conservatively-sized, diversified beta basket does exactly what it is supposed

to do, it survives, but survival is the wrong objective: the two-step needs the path to *arrive* at a target, and a +1%/yr drift does not arrive inside a deployable window at any cost. Concentrating into the strongest legs only converts the problem into a directional bet on the operator's regime call, positive when the call is right in hindsight and indistinguishable from zero unconditionally, and no look-ahead-free rule we can build supplies the call cheaply enough to survive the whipsaw. This is the barrier-native restatement of a fact the program already held on other axes: conditioning a beta stream on a regime state does not add a harvestable return, it just relabels realized drift as timing. That is why the finding is consistent with, and does not disturb, the standing *s1-dynamic-conditioning* belief and the 0/16 base rate.

None of this says beta harvesting is worthless in general; in an unconstrained account a +1%/yr diversified carry is a perfectly real, if modest, return. What it says is narrower and deployment-specific: *under a first-passage barrier with a fee*, a confirmed beta stream does not pay, because the barrier rewards drift-to-target and punishes the one thing beta gives you cheaply, which is variance without enough drift.

7 Limitations

Five honest caveats. First, the price paths are futures proxies for FTMO CFDs, and the fee, swap, and event-calendar inputs are placeholders pending a human-in-the-loop source freeze; the diversified verdict is insulated from this by its cost-independence, but the concentrated EV numbers would move with real costs (and the unconditional EV, already near zero, could move either side of it). Second, the diversified gate numbers are a *provisional single slice*: the full gate-grade adjudication (walk-forward per-fold edge-CI and rolling-2022 origins) is still pending, so the STOP rests on the structural, cost-independent drift ceiling rather than on a completed walk-forward gate. Third, the concentrated {NQ, GC} basket is a post-hoc, single-composition probe, not a pre-registered strategy; it is an upper bound and nothing more, and its low driftless floor (0.04 to 0.06) is a pinned-vol-target statement that rises with the vol target (to about 0.12 in the bear at twice the size), so “the barrier geometry alone almost never passes” is scoped to the pinned config. Fourth, the 200-SMA gate result is *provisional*: it uses realized additive equity paths rather than the full block-bootstrap barrier engine, does not charge the gate-flip transaction cost (which makes the causal gate look better than it is, so the conclusion is conservative), and tests a single mechanical rule; a discretionary regime call that reliably beats a lagging trend filter is a high, un-backtestable claim, but it is the only door the falsification leaves open. Fifth, the breadth is deliberately narrow, one firm's two-step, one account size, a small set of liquid legs; that is the right scope for a deployment decision but a limit on external validity.

8 Conclusion

A prop-firm challenge is a first-passage race, and “harvest beta, size it small” loses that race in three different ways. The pre-registered diversified basket hits a structural +1%/yr drift ceiling and cannot reach the target even for free; a maximally hindsight-favorable concentration into the two strongest legs is positive only on a declared bull window and, unconditionally, a coin-flip that straddles zero and never clears the fee; and a look-ahead-free mechanical regime gate whipsaws the bull edge to zero while deepening the drawdown. The whole positive expectation the recipe seems to offer lives in regime foreknowledge that no honest, causal rule supplies. The transferable part is the method: a first-passage simulator that runs the real barriers, a driftless null regenerated inside the engine to tell realized drift from geometry, a cost-independence check that names the binding wall, and a HARKing antidote that swaps a hindsight window for a causal rule and watches the edge evaporate.

The deployment consequence is blunt: the challenge fee, not the in-trade notional, is the true risk unit, and no forward-testable version of the beta-harvest recipe justifies spending it. The standing verdict is unchanged and now better argued: NOT-PROVEN.

Reproducibility and firewall. All statistics derive from the frozen run artifacts `runs/ftmo_v2/nqgc_basket/{nqgc_basket_regimes,nqgc_sma_gate_provisional}.json`, `runs/ftmo_v2/gates/provisional_single_slice.json`, and the pinned diagnosis in `docs/prop/ftmo-v2-provisional-findings.md`; figures and the canonical data/`numbers.json` are produced by `scripts/make_figures.py`. This is a *prop-deployment*-context write-up (ADR 0064, section 3): it writes nothing to `docs/findings/` or `docs/beliefs/`, does not move the 0/16 research base rate, and the three beta recipes remain derived and NOT-PROVEN. Directions and gates were fixed a priori in a frozen pre-registration; the concentration probe is labeled post-hoc throughout. Numbers are frozen as of July 2026.

Appendix A: statistics of record

The tables below are extracted from the frozen run artifacts (and, for the R1 drift and survival rows, explicitly hard-set from the pinned findings note) via `data/numbers.json`; they are the single source of truth for every number in the body.

Result	Statistic	Value
R1 diversified (vt 0.09)	drift, gross / flat 4 bps	+1.00% / +0.83% per yr
	drift, half / placeholder swap	−1.94% / −4.87% per yr
	$P(\text{pass} \wedge \text{retain}) @ 63\text{d} / 126\text{d}$	0 / 0
	edge @ 63d / 126d	0 / 0
	$P(\text{pass}) @ \text{infinite window}$	0.02
	2022 realized: final / worst DD	−4.1% / −6.1%
	full 2010–26: final / worst DD	+8.5% / −6.3%
R2 concentrated {NQ,GC} (vt 0.10)	bull: P, driftless, edge	0.763, 0.036, +0.726
	bull: business EV	+\$681
	full: P, driftless, edge	0.313, 0.036, +0.276
	full: business EV	−\$28
	bear: P, driftless, edge	0.026, 0.058, −0.031
	bear: business EV	−\$489
R2 sensitivity (vt 0.20)	full / bull / bear EV	+\$24 / +\$272 / −\$348
R3 200-SMA gate (vt 0.10, prov.)	ungated: drift / maxDD	+2.61% / −29.5%
	hindsight bull: drift / maxDD	+10.14% / −13.2%
	gated: drift / maxDD	+0.12% / −39.8%
	gate switches / % in market	27 / 79%

Appendix B: pre-registration and challenge trail

The diversified basket, its vol target and leg set, the hazard-policy levers, the evaluation and funded-horizon pins, and the four gates (G1 edge, G2 2022 survival, G3 business EV, G4 terms-of-service conformance) are pre-registered, frozen before any number, in `docs/prop/ftmo-v2-preregistration.md` (v2, itself revised by an admissible-with-changes pre-commit challenge). The provisional build and the G1 drift-ceiling diagnosis are recorded in `docs/prop/ftmo-v2-provisional-findings.md`; the concentrated-bet deployment rulebook and the 200-SMA falsification (Nachtrag A) in `docs/prop/ftmo-v2-live-deployment.md`. The multi-round derivation saga (NO-GO to NOT-PROVEN to material-viable to refuted, across three same-model verdict flips and two cross-model passes) is ADR 0064, Nachträge 1 through 7. The independent adversarial challenge for the concentration probe is logged at `runs/challenges/ftmo_v2_nqgc_2026-07-03.md`, and the cross-model (gpt-5.5 / codex) pass at `runs/challenges/ftmo_v2_nqgc_codex_2026-07-03.md`; both returned *sound with caveats*, beta-not-alpha confirmed, no verdict-flipping defect.